
CY 646: Advanced NMR Spectroscopy- A problem based approach

Course Code	CY 646
Course Name	Advanced NMR Spectroscopy- A problem based approach
L-T-P-C	3-0-0-3
Prerequisites	BSc (with Chemistry as one of the subject)
Intended for	PG
Distribution	Elective Course
Approval	8th Senate (Oct 2015)

Course Contents

- **Nuclear Magnetic Resonance Spectroscopy**
The NMR Phenomenon, The spinning nucleus, the effect of an external magnetic field, Precessional motion, Precessional frequency, Energy transitions. [2 Lectures]
- **Theory of Nuclear Magnetic Resonance**
Chemical Shift and its Measurement, internal standards, the NMR spectrometer, units used in NMR spectroscopy, Factors Influencing Chemical Shift, Electronegativity shielding and deshielding, anisotropic effects, Correlation Data for Proton NMR Spectra, Use of correlation tables, Influence of restricted rotation. Choice of solvent for proton NMR spectra, Solvent shifts concentration and temperature effects hydrogen bonding, Integrals in Proton NMR Spectra. [3 Lectures]
- **Symmetry and Topicity**
Homotopicity, Enantiotopicity, Diastereotopicity, Chemical Equivalence, Magnetic Equivalence. [1 Lectures]
- **Through-Bond Effects**
Spin-Spin (J) Coupling & Origin of J-Coupling Skewing of the Intensity of Multiplets, Prediction of First-Order Multiplets, The Karplus Relationship for Spins Separated by Two Bonds, The Karplus Relationship for Spins Separated by Two Bonds, Long Range J-Coupling, Decoupling Methods, One-Dimensional Experiments Utilizing J-Couplings, Two-Dimensional Experiments Utilizing J-Couplings, Homonuclear. [2 Lectures]
- **Two-Dimensional Experiments utilizing J-Couplings**
COSY, HMQC, HSQC, HMBC, TOCSY, INADEQUATE, Heteronuclear Two-Dimensional Experiments Utilizing J-Couplings. [4 Lectures]
- **Through-Space Effects**
The Nuclear Overhauser Effect (NOE)
The Dipolar Relaxation Pathway, The Energetics of an Isolated Heteronuclear Two-Spin System, Decoupling One of the Spins in a Heteronuclear Two-Spin System, Rapid Relaxation via the Double Quantum Pathway, A One-Dimensional Experiment Utilizing the NOE, Two-Dimensional Experiments Utilizing the NOE, NOESY and ROESY. [4 Lectures]
- **Strategies for Assigning Resonance to Atoms within a Molecule**
Prediction of Chemical Shifts, Prediction of Integrals and Intensities, Prediction of ^1H Multiplets, good Bookkeeping Practices, Assigning ^1H Resonances on the Basis of Chemical Shifts, Assigning Resonances on the Basis of Multiplicities, Assigning ^1H Resonances on the Basis of the gCOSY Spectrum, The Best Way to Read a gCOSY Spectrum, Assigning ^{13}C Resonances on

the Basis of Chemical Shifts, Pairing ^1H and ^{13}C Shifts by Using the HSQC/HMQC Spectrum, Assignment of Nonprotonated ^{13}C 's on the Basis of the HMBC Spectrum, Variable Temperature NMR techniques. ^{19}F and ^{31}P NMR. [4 Lectures]

- **Strategies for Elucidating Unknown Molecular Structures**

Initial Inspection of the One -Dimensional Spectra Good Accounting Practices, Identification of Entry Points, Completion of Assignments. [1 Lectures]

- **Simple Problems**

2-Acetylbutyrolacton, α -Terpinene, (1R)-endo-(+)-Fenchyl Alcohol in CDC13, (-)-Bornyl Acetate, N-Acetylhomocysteine Thiolactone, Guaiazulene, 2-Hydroxy-3-Pinanone, (R)-(+)- Perillyl Alcohol, 7-Methoxy-4-Methylcoumarin, Sucrose and some more interesting molecules from literature if time permits. [8 Lectures]

- **Complex Problems**

Longifolene, (+) Limonene, l-Cinchodine, (3aR)-(+)- Sclareolide, (-)-Epicatechin, (-)- Eburnamonine, trans-Myrtanol, cis-Myrtaol, Naringenin, (-) Ambroxide and some more complex molecules from literature if time permits. [10 Lectures]

Text Books

1. Harald Gunther, **NMR Spectroscopy**, 2nd Edition, Wiley Publishers, 1995.
2. Jeremy K.M. Sanders, Edwin C. Constable, Brian K. Hunter and Clive M. Pearce, **Modern NMR Spectroscopy: A workbook of Chemical Problems**, 2nd Edition, Oxford University Press, 1993.
3. Jeffrey H. Simpson, **Organic Structure Determination, using 2-D NMR Spectroscopy, a problem based approach**, Academic Press, 2012.

References

1. Erno Pretsch, Philippe Buhlmann, Martin Badertscher, **Structure Determination of Organic Compounds, Tables of Spectral Data**, Springer, 2009.
2. Robert M. Silverstein, Francis X. Webster, David Kiemle, **Spectrometric Identification of Organic Compounds**, 7th Edition, John Wiley & Sons, 2005.
3. William Kemp, **Organic Spectroscopy**, 3rd Edition, Palgrave Publishers Ltd.